effects. Fire effects are controlled by gizmo helper objects, which are used to place and scale the effect. Multiple fire effects can be added, so their order in the Effects list becomes important. Fire does not create light or shadows, so it is common to create a light that simulates this part of the effect. The effect is controlled by a gizmo helper, and light is provided by an omni light.

## 5.6 Mental ray renderer

Mental ray for 3ds Max is an excellent renderer that has become a standard at many studios around the world. Mental ray can render scenes by using its own scanline algorithm (not to be confused with 3ds Max's default scanline renderer) and it can also ray trace. Mental ray can be more robust than 3ds Max's scanline renderer when it comes to creating highly realistic effects. Global Illumination allows mental ray to simulate the way light bounces off diffuse surfaces, much like Advanced Lighting within the scanline renderer. This can create a much softer and more-realistic scene. Caustics simulate the way light reflects or passes through complex objects and can simulate materials such as glass, water, and reflective metals. Mental ray also has a robust library of material types that can simulate surfaces such as glass, skin, and car paint.

Mental Ray diamonds

3ds Max has fairly tight integration with mental ray, meaning much of it is invisible to the average user. Shaders, lights, and cameras from 3ds Max transfer over to mental ray with little effort. For additional control, cameras, lights, objects, and shaders all have mental ray attributes that become active when mental ray is used as the renderer. Additional mental ray rendering features such as Global Illumination and Caustics can be used on most 3ds Max scenes as well. Basic controls over the mental ray renderer are located on the Render Scene window's Renderer tab.

### 5.6.1 Global illumination

Global illumination is a rendering method used to create highly realistic lighting and shading. It simulates the actual scattering of photons of light around the scene. Although it is computationally expensive, the results are